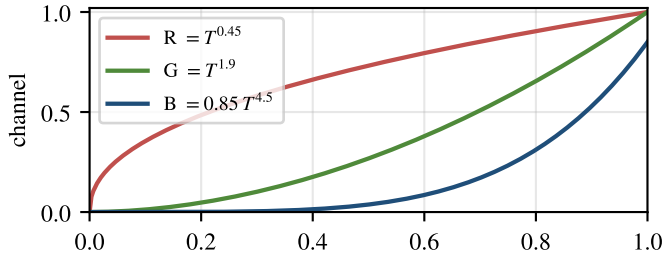
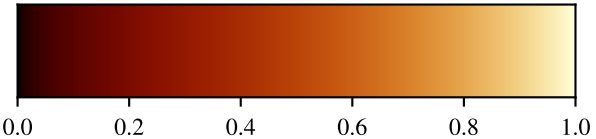
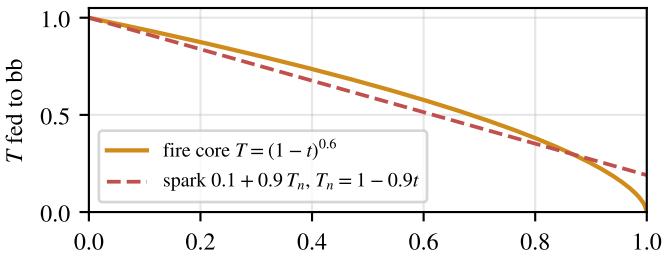
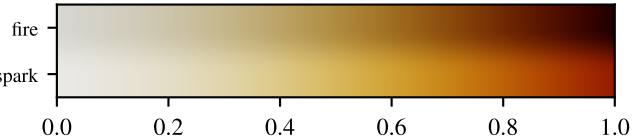

(a) shader ramp bb(T): red first, then green, then blue
channel
1.0
0.5
0.0
R = T^0.45
G = T^1.9
B = 0.85 T^4.5
0.0
0.2
0.4
0.6
0.8
1.0
temperature T (0 = cold, 1 = core)


0.0
0.2
0.4
0.6
0.8
1.0
bb(T): black, deep red, orange, yellow, white-yellow


(b) temperature over a particle's life
T fed to bb
1.0
0.5
0.0
fire core T = (1 − t)^0.6
spark 0.1 + 0.9 T_n, T_n = 1 − 0.9t
0.0
0.2
0.4
0.6
0.8
1.0
normalised life t = age / life


fire
spark
0.0
0.2
0.4
0.6
0.8
1.0
emitted colour over life (bb × intensity, tone-mapped)